

The background features several overlapping circles. Some are filled with dark blue diagonal stripes, while others are solid dark blue. The circles are positioned in the corners and along the edges of the frame, creating a modern, geometric aesthetic.

# **MATLAB- Integrated, Arduino-Based Environment Mapping system**

# Overview

**01 Introduction**

**02 Objective**

**03 Components**

**04 Methodology**

**05 Working**

**06 Significance of  
MATLAB**

**07 Work done**

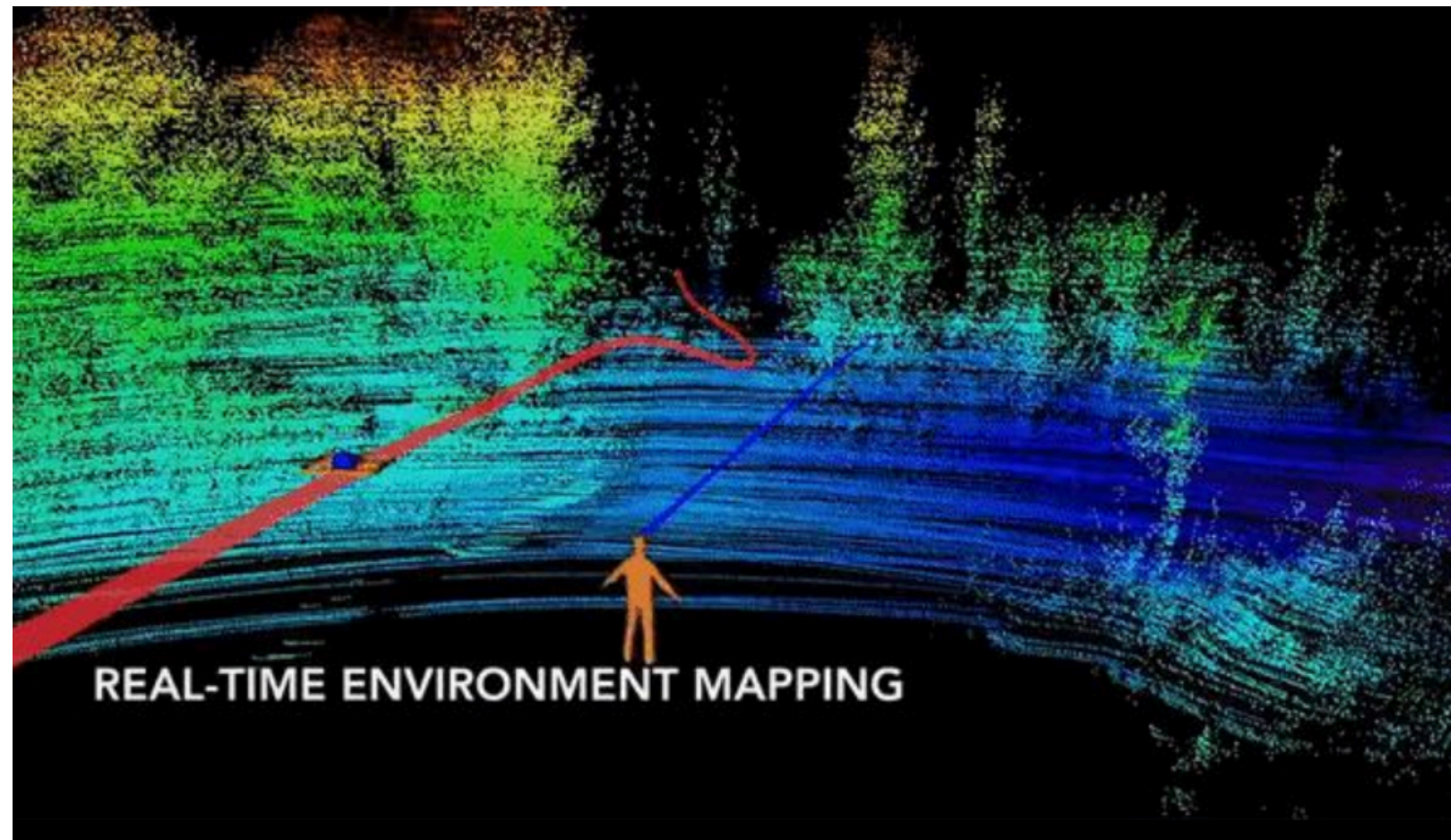
**08 Conclusion**



# Introduction

Modern intelligent systems require accurate perception of their surroundings for automation, navigation, and decision-making. This project combines Arduino-based sensing with MATLAB processing to develop a smart environment mapping system capable of real-time data acquisition, visualization, and analysis. By integrating ultrasonic sensing, sensor fusion, and graphical mapping techniques, the system improves object detection accuracy and enhances environmental awareness for robotic and automation applications.

# Environmental Mapping



Environment mapping is the process of collecting spatial information from sensors and converting it into a structured visual representation of the surroundings. In this project, ultrasonic sensors continuously measure distances to nearby objects, while MATLAB processes the collected data to generate a real-time map of the environment.



# Objective

To Develop a MATLAB -Integrated Arduino System for real time environmental data collection and to represent data in clear graphical format

To Improve understanding of object positions and surroundings through visual maps





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# Problem Statement

Traditional environment monitoring and mapping methods are often expensive, complex, and require advanced hardware. There is a need for a low-cost and efficient system capable of detecting obstacles and creating environmental maps in real time. Existing systems may lack flexibility and easy visualization of sensor data. Therefore, a MATLAB-integrated Arduino-based environment mapping system is proposed to collect distance data using sensors, process it through Arduino, and visualize the surroundings using MATLAB for better analysis and understanding





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# Components Used

- Arduino Uno Microcontroller- Acts as the central control unit for sensor interfacing, data acquisition, and communication with MATLAB.
  - HC-SR04 Ultrasonic Sensor- Measures the distance of nearby objects using ultrasonic wave reflection for obstacle detection and mapping.
  - Servo Motor Mechanism-Rotates the ultrasonic sensor across multiple angles to perform wider environmental scanning and area coverage.
  - Breadboard and Connecting Wires-Provide temporary circuit assembly, stable electrical connections, and flexible hardware integration.
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- Computer with MATLAB Software- Processes sensor readings, performs real-time data analysis, and generates graphical environment maps.
  - Power Supply Module- Ensures stable voltage and reliable operation of Arduino and peripheral components during continuous scanning.
  - Serial Communication Interface-Enables efficient transfer of sensor data between Arduino hardware and MATLAB visualization tools.
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# Methodology

## 1. Hardware Initialization

Arduino Uno initializes the ultrasonic sensor and servo motor, preparing the system for environmental scanning.

## 2. Sensor-Based Data Acquisition

The ultrasonic sensor continuously measures distances while the servo motor rotates through predefined angles for wider coverage.

## 3. Real-Time Data Transmission

Collected sensor data is transmitted from Arduino to MATLAB using serial communication protocols.



#### 4. Data Processing and Filtering

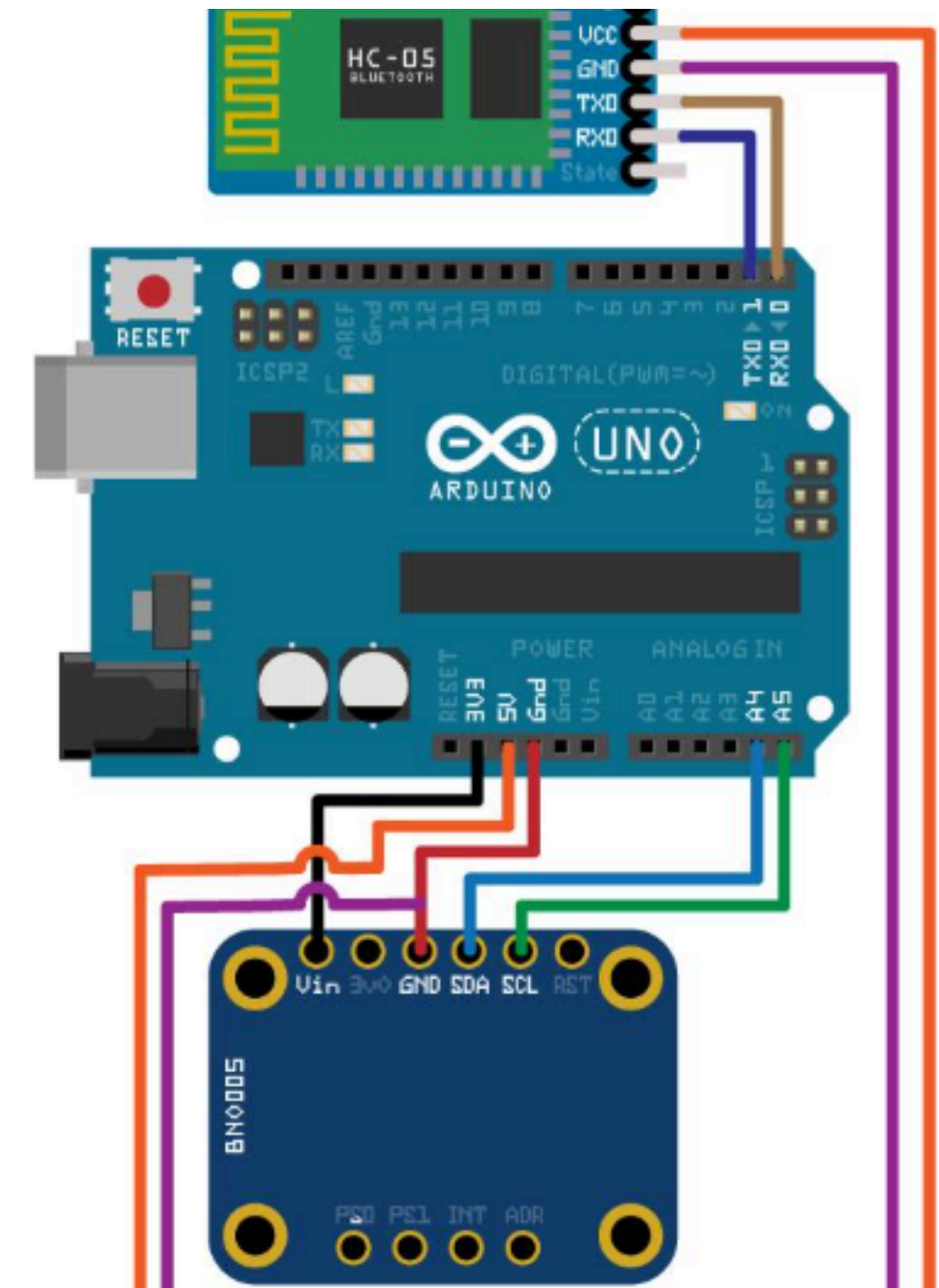
MATLAB processes incoming readings, removes noise, and converts raw sensor values into meaningful spatial information.

#### 5. Environment Mapping and Visualization

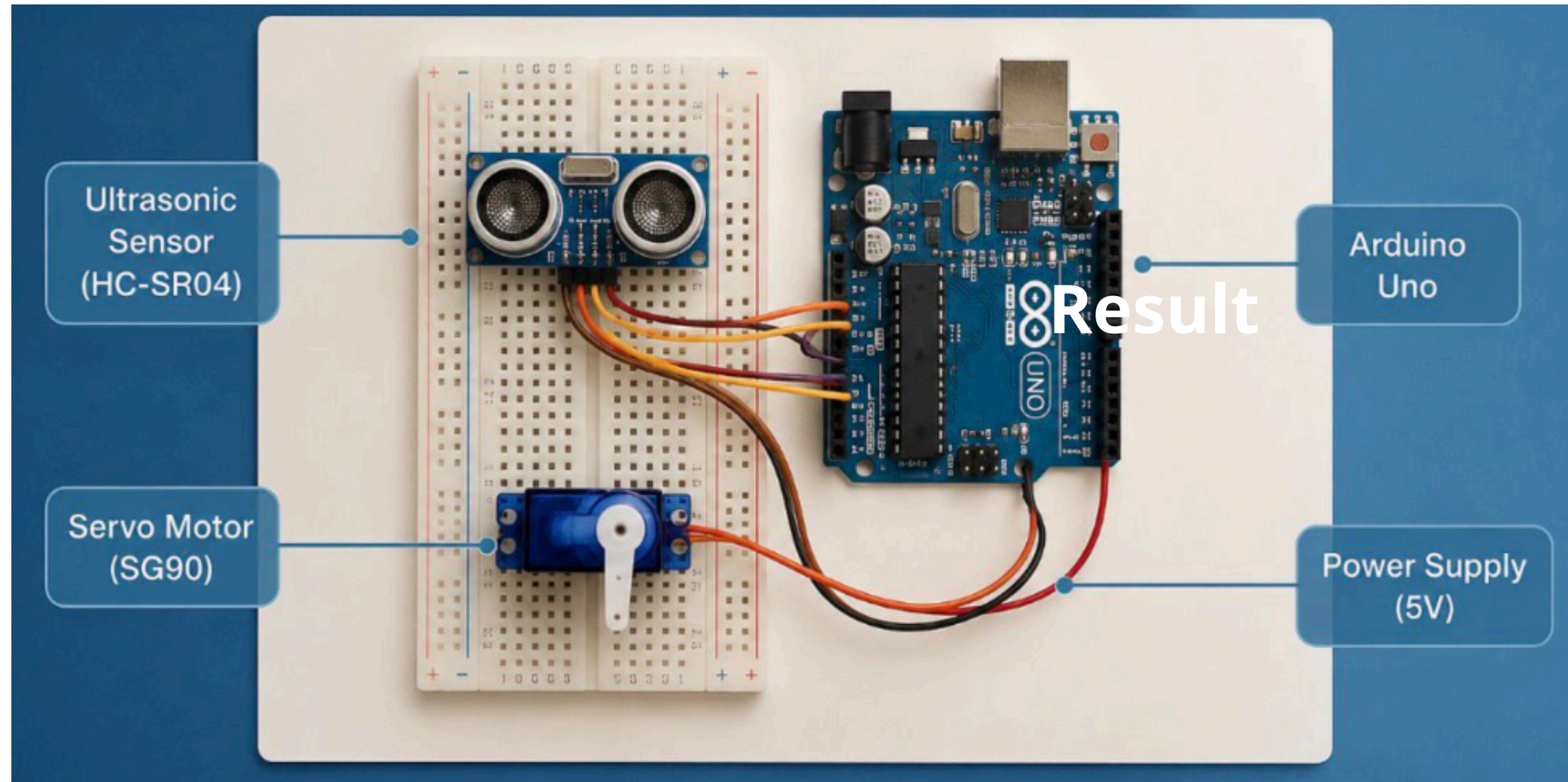
Processed data is displayed graphically as a real-time environmental map showing object positions and obstacle locations.

#### 6. Continuous Monitoring and System Optimization

The system continuously updates the map for dynamic environments while improving scanning accuracy and response speed.



# Connection Diagram

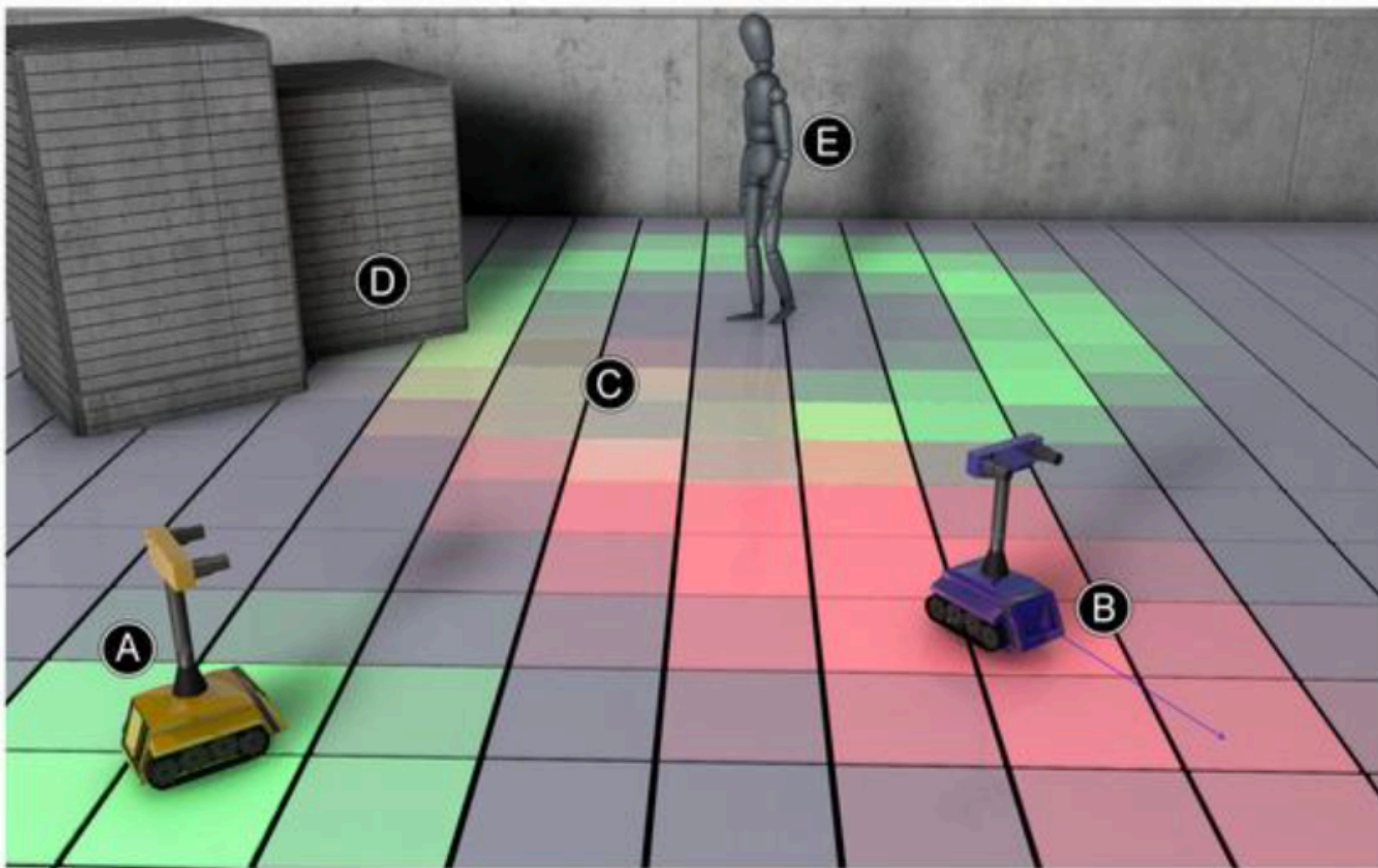




# Working

- \* The sensor rotates and scans the surroundings at different angles like  $0^\circ$ ,  $30^\circ$ ,  $60^\circ$ , and  $90^\circ$ .
- \* At every angle, the sensor measures the distance between the sensor and nearby objects.
- \* If no object is present, the distance value remains very high or unstable.
- \* When an object is detected, the measured distance suddenly decreases because the signal gets reflected back.
- \* A group of nearby distance points at similar angles confirms the presence of a real object.
- \* The collected data is then processed and displayed graphically using MATLAB for better visualization and analysis.

# Significance of using MATLAB



- \* MATLAB provides powerful tools for processing and analyzing sensor data efficiently.
- \* It helps in generating real-time graphical visualization of scanned objects and surroundings.
- \* MATLAB improves accuracy in object recognition and mapping applications.
- \* Large amounts of sensor data can be stored, filtered, and analyzed easily.
- \* It supports advanced algorithms for robotics, automation, and autonomous navigation systems.
- \* MATLAB also simplifies debugging, simulation, and performance testing of the system.

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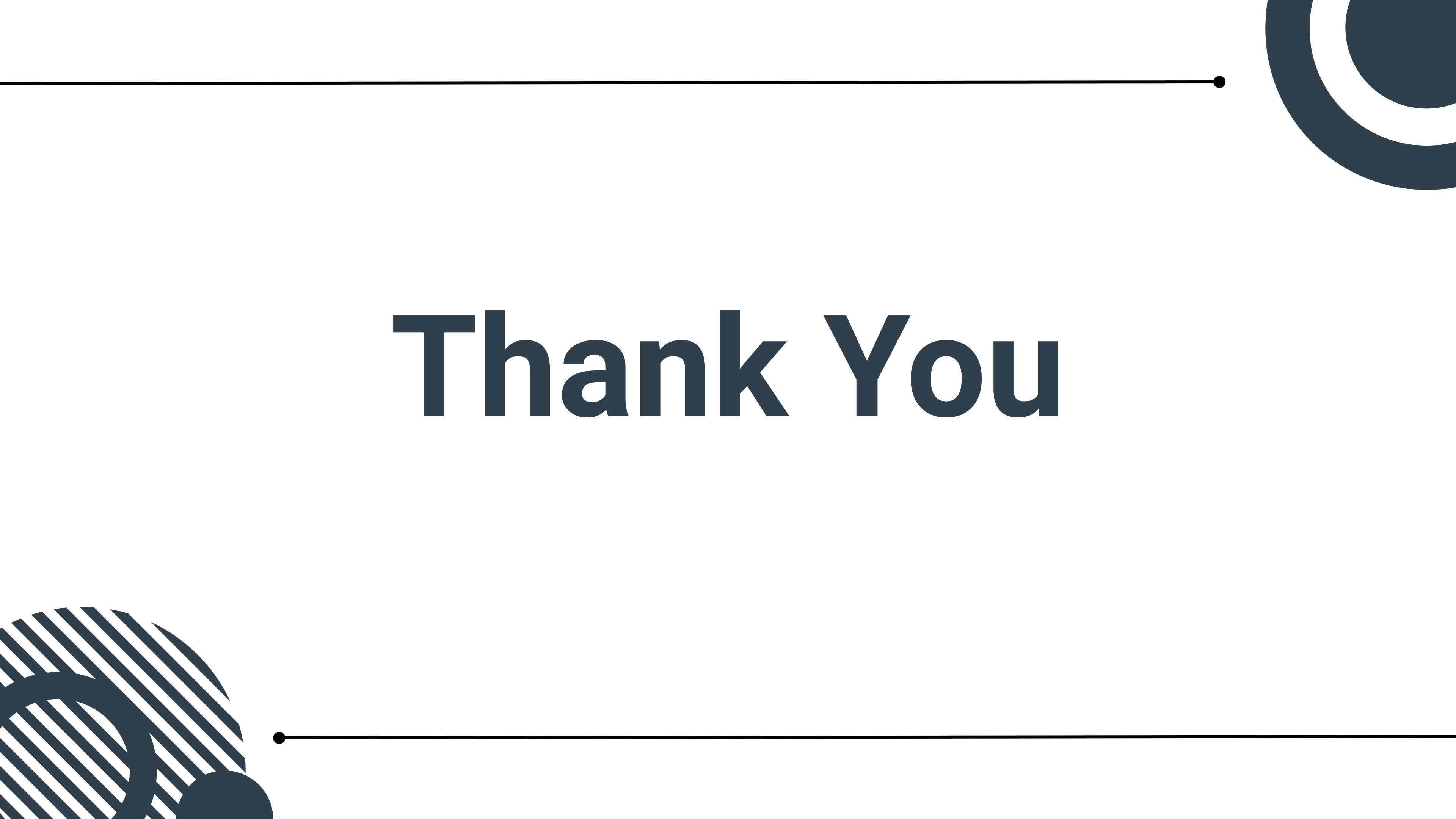
# Key Applications

- Used in autonomous vehicles for navigation and obstacle detection.
  - Helpful in robotics for path planning and intelligent movement.
  - Applied in industrial automation and smart manufacturing systems.
  - Used in surveillance and security systems for environment monitoring.
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# Conclusion

This project successfully demonstrates the integration of Arduino and MATLAB for real-time object detection and environmental mapping. The system efficiently scans surroundings, detects nearby obstacles, and visualizes sensor data with improved accuracy and reliability. By combining hardware control with advanced data processing, the model provides a scalable foundation for intelligent automation and robotic applications.



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# Thank You

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